

DRIFT MONITORING PROGRAM

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BACKGROUND

- Various UCD departments are developing validated “data assets”
- A data asset is data representing a specific concept
  - Hospital length of stay
  - Duration on mechanical ventilation
  - Number of creatinine labs ordered
- SQL code generates a data asset from EHR and operational data base fields
- Assets are validated through chart review
- Creating a fully validated data asset can be time consuming



DRIFT MONITORING

- Important that assets used for patient care and hospital operations remain correct and reliable
- Circumstances can change over time
  - Changed circumstances – the environment or trend have changed (e.g., COVID)
  - New data fields are added or sun-setted
  - Lab tests change (e.g., vendor change) - units change, detection limits
  - Something breaks
- For prediction models
  - Prevalence can change
  - Population characteristics can change
  - Data availability can change
  - Input data can change
  - Relationship between model features and outcome can change

DEVELOPING DRIFT MONITORING PROGRAM

- Desire to develop “standardized” drift monitoring program
  - Automated or easily routinely implementable by analyst
  - Generally applicable to all data assets
    - Recognize we need to consider different data types (categorical, counts, continuous)
- Needs to capture
  - “outliers” – something broke
  - abrupt drift – change in how data are reported
  - gradual drift – environment/patient population is changing over time

NUMERIC DATA TEST CASE
TIME ON MECHANICAL VENTILATION

- Data from April 2011 to April 2022
  - Pre and post-COVID
- Encounter-level data: 21,819 encounters
- Time on mechanical ventilation in hours
- Date indexed to start of mechanical ventilation

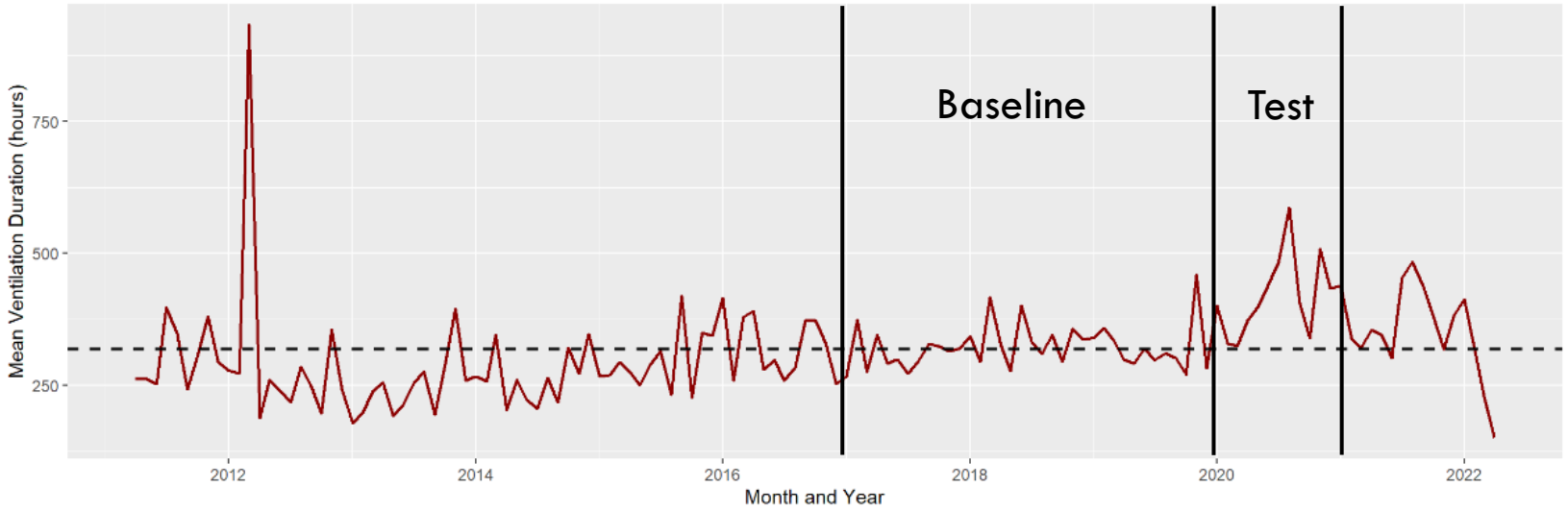
PROPOSING THREE-COMPONENT APPROACH

- Outlier detection
  - Monthly assessment of summary metrics to identify extreme events
- Distributional drift
  - 3-month rolling window evaluating shifts in the distribution of mechanical ventilation
  - Less volatile than monthly assessment
  - Considers entire distribution rather than summary statistics
- Annual trends over time
  - Annual look back evaluating change in longer term trend
  - Monthly and 3-month assessments might not capture longer term changes
- Program specifics are still being refined

COVID NATURAL EXPERIMENT

- For illustration purposes
  - Baseline: 2017 – 2019
  - Test period: 2020
- 2017-2019
  - Stable MV durations
- 2020
  - Increased MV durations

Mean monthly mechanical ventilation duration



OUTLIER DETECTION

OUTLIER DETECTION

- Evaluate monthly median, 25th, 75th, minimum, maximum, and IQR
  - Considering all of these metrics helps capture a diversity of potential shifts and changes
- Tukey fences to identify extreme events
  - Tukey limits less influenced by skewed distributions than mean ± 3SD
  - Can be applied to any numeric variable without consideration of distribution
- Establish evaluation limits of Tukey fences from Baseline period using monthly values

OUTLIER DETECTION FOR 2020

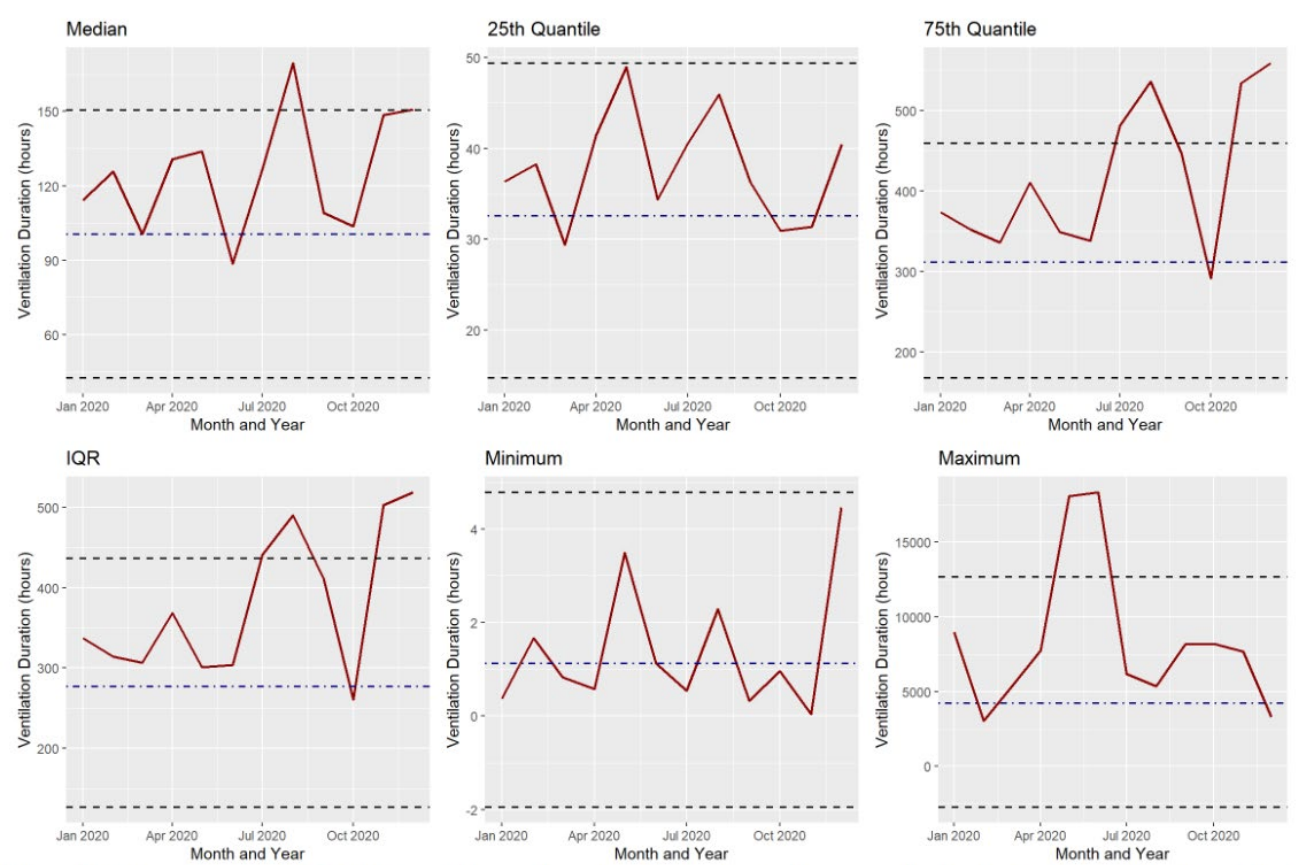


Figure 4.4: Tukey fences (+/- 1.5 IQR) for monthly mechanical ventilation duration (hours) metrics for the Test Year 2020 . Gray dotted dashed line is median from baseline (2017-2019) period. Black dashed lines are quartiles +/- 1.5IQR.

- Tukey fences would trigger alert
  - August 2020 based on median, 75th quantile, and IQR
  - May & June 2020 based on maximum
  - December 2020 based on 75th quantile and IQR
- These months align with COVID surges so expected
- Fence limits can be refined for desired sensitivity

DISTRIBUTIONAL DRIFT ASSESSMENT

DISTRIBUTIONAL DRIFT

- Considered Kolmogorov-Smirnov Test
  - KS Test evaluates the null hypothesis that two distributions are the same
  - Concern about rejecting null for small differences because of large sample sizes
- Equivalence testing could be better framework
  - Null is that two distributions differ by at least X amount and rejecting null indicates that two distributions are equivalent within the specified margin (X)
- Lei & Olson (2019) Evaluating statistical methods to establish clinical similarity of two biologics. J. Biopharm Statistics 20: 62-74

OVERLAP COEFFICIENT TEST

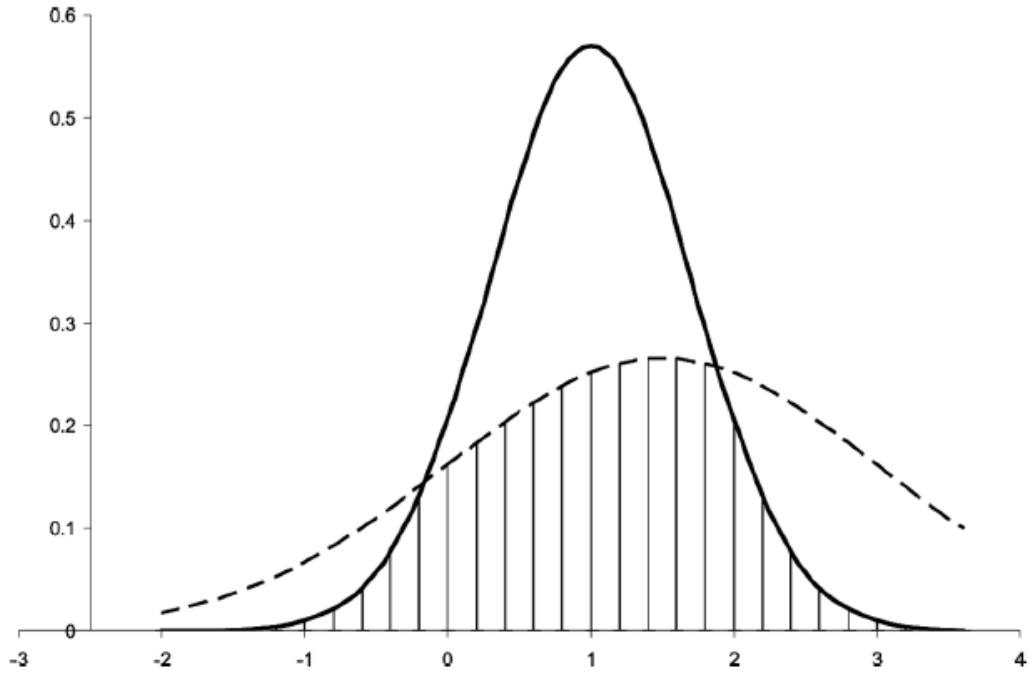


Figure 1 An illustration of parametric overlap coefficient method.

- Uses percentage of overlap of distributions as metric (OC)
  - OC = 1 complete overlap
  - OC = 0 no overlap
- Equivalence testing framework
  - Ho: OC(%) < *m* vs. Ha: OC(%) ≥ *m*
  - *m* is the minimum acceptable overlap (i.e. equivalence margin)

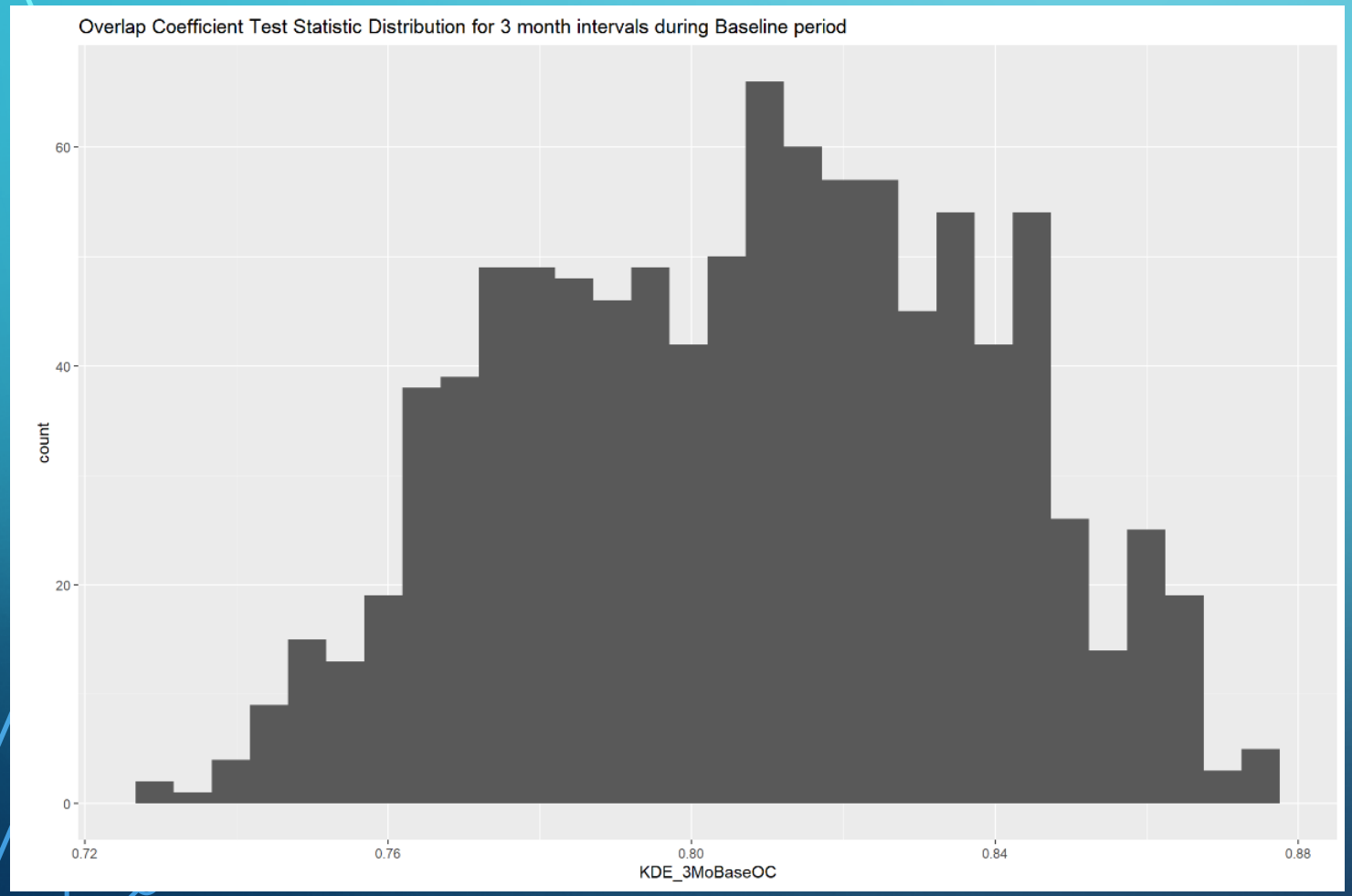
OVERLAP COEFFICIENT TEST IMPLEMENTATION

- Use kernel density estimation to estimate baseline and test set distributions and estimate overlap (overlapping package in R)
- Draw B bootstrap sample from baseline and test set and calculate overlap for bootstrap samples
- Calculate the lower 95% confidence limit from bootstrap samples as

$$LL(95) = \widehat{OC} - 1.645 \cdot SD(OC)$$

- Reject Ho if LL(95) > *m* and conclude equivalence within margin of *m*

APPLICATION TO VENTILATION DATA ESTABLISHING EQUIVALENCE MARGIN



- Calculated OC for 1000 bootstrapped 3-months from baseline and calculated OC for the 3 selected months with the remaining 33 months

Median [min, max] = 80.9 [73.1, 87.7]

Mean +/- SD = 80.8 +/- 3.0

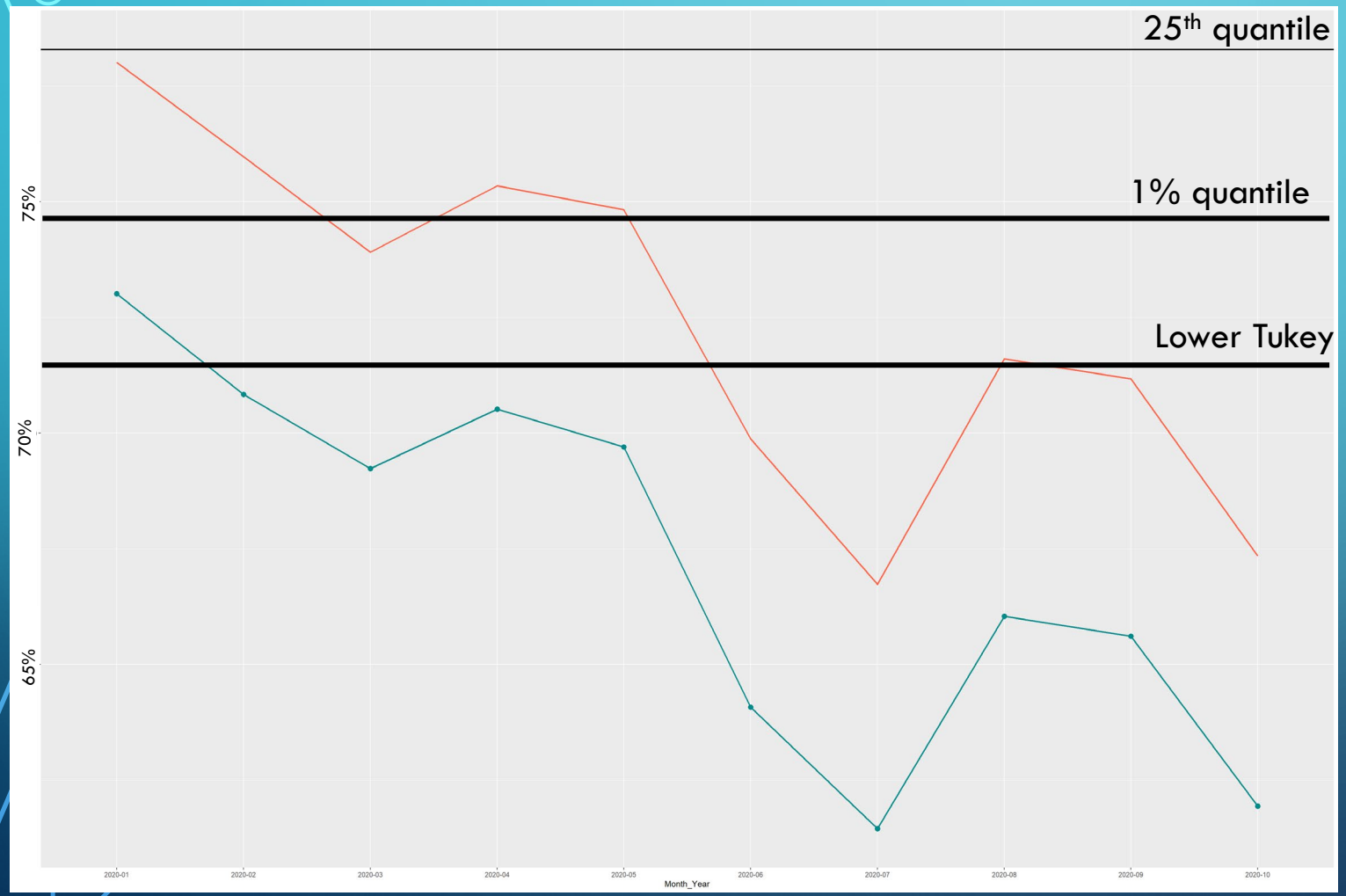
(-2SD = 74.7)

1st quantile = 74.4

5th quantile = 75.9

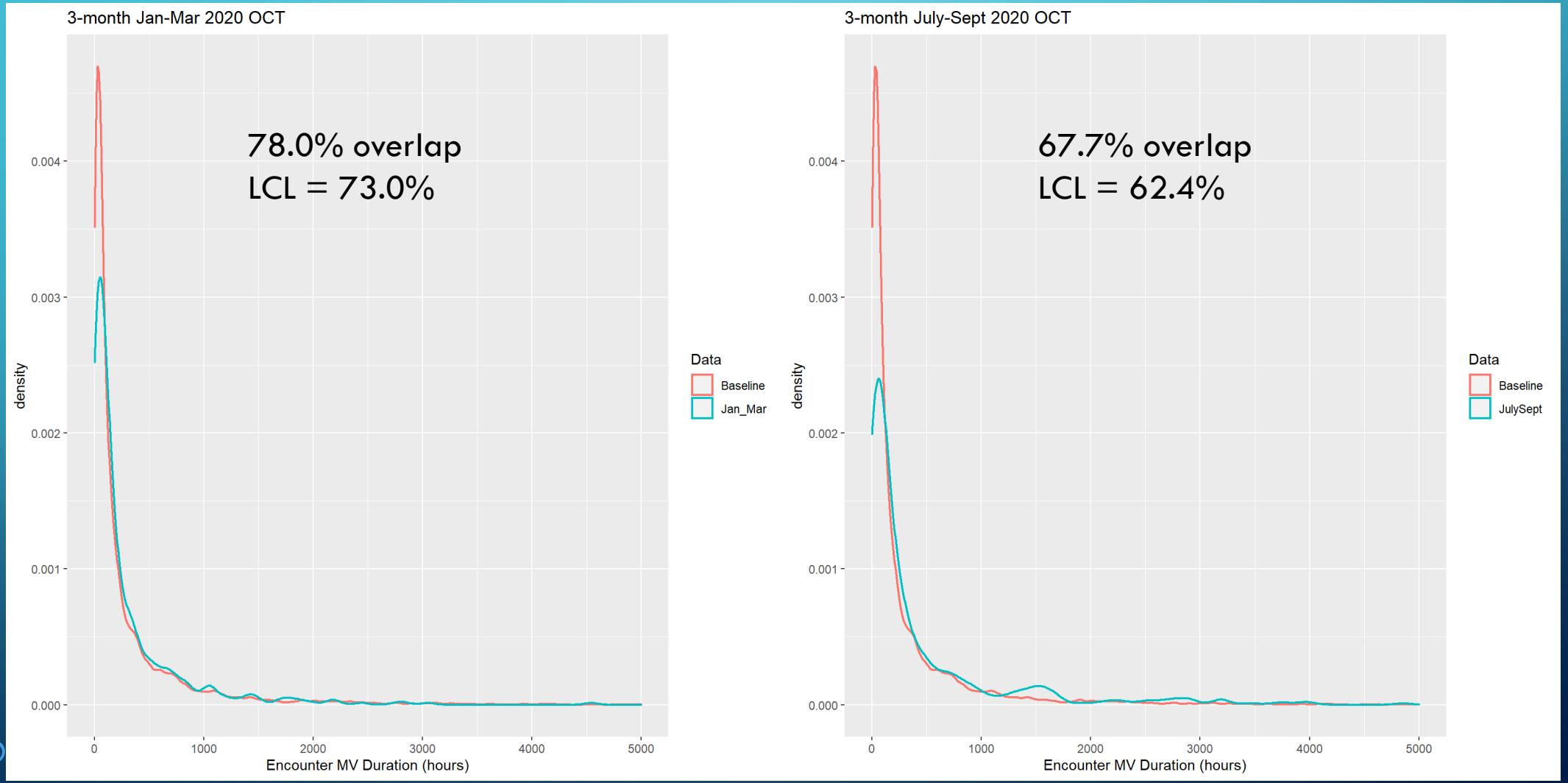
Lower Tukey fence = 71.0

APPLICATION TO VENTILATION DATA EVALUATING 2020 DATA

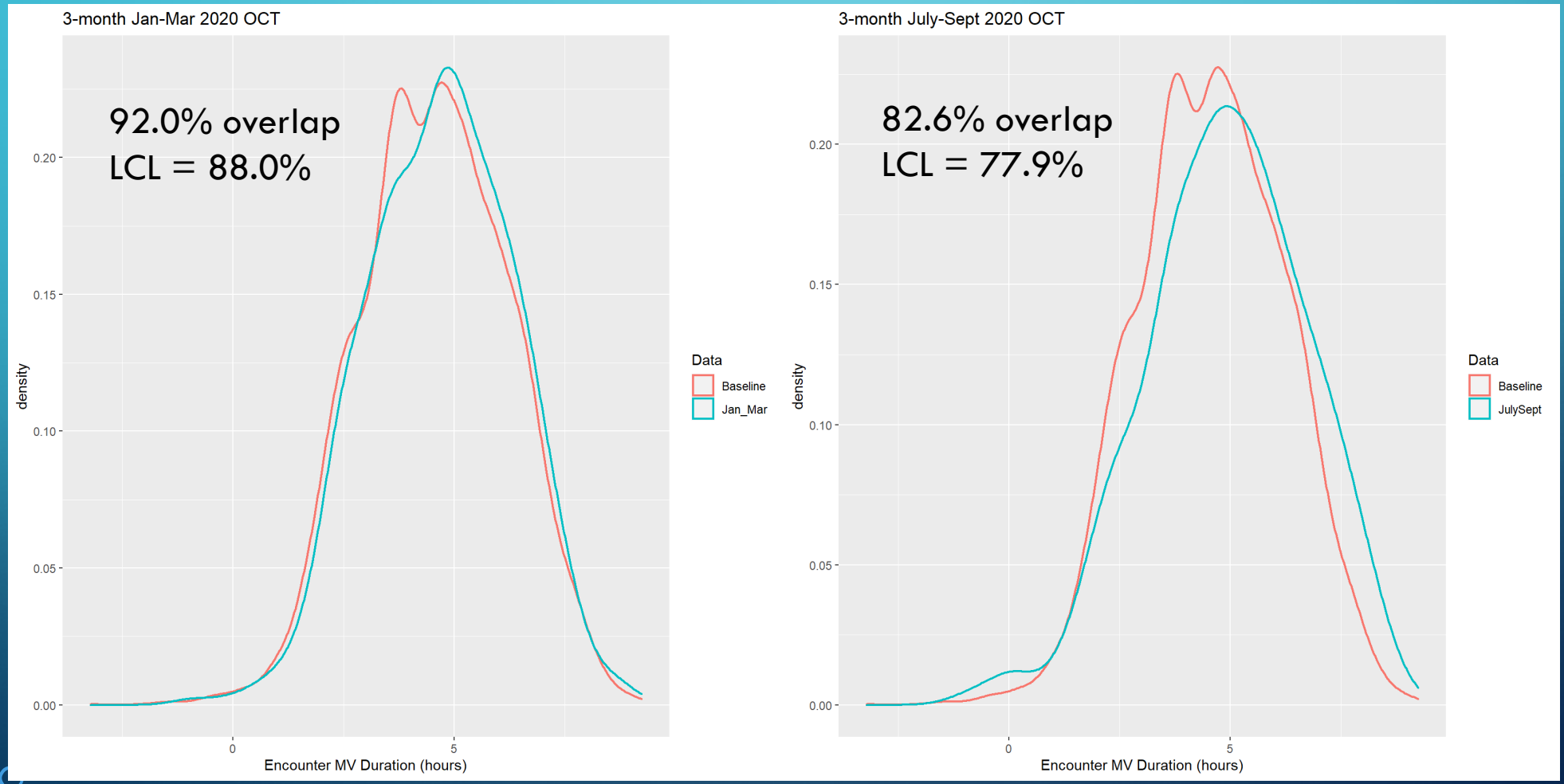


- Evaluated rolling 3-month windows
- Estimated LCL(95) from 1000 bootstraps
- Plotted OC estimates and LCL(95) versus alternative margins
- Indications of shift in distribtion

MINIMUM AND MAXIMUM OVERLAP MONTHS 2020 VS. BASELINE



MINIMUM AND MAXIMUM OVERLAP MONTHS FROM BASELINE FOR LOG TRANSFORMED DATA



OVERLAP COEFFICIENT TEST CONSIDERATIONS

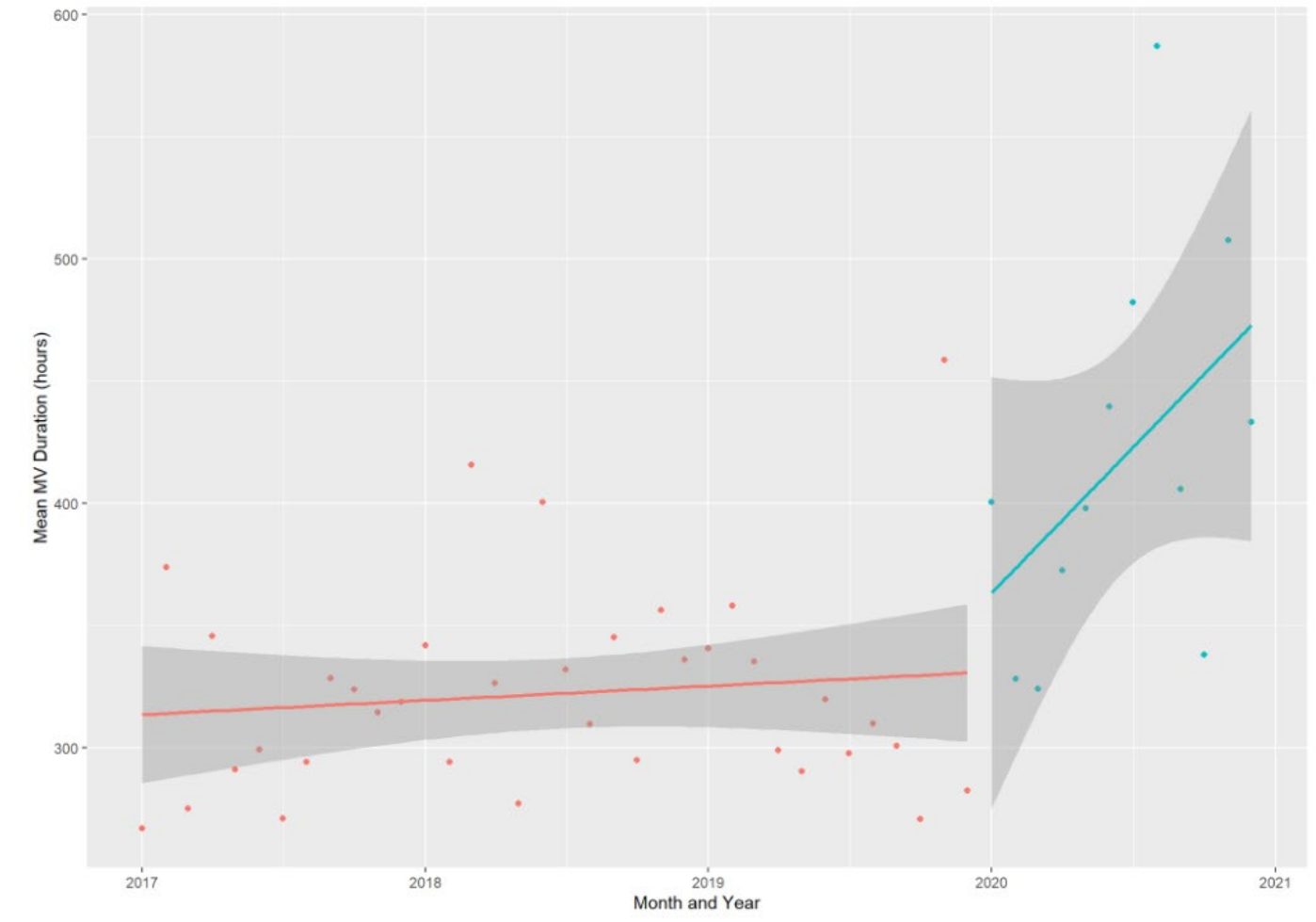
- 3 – month rolling window is probably a good compromise
- Log transforming strongly skewed variables may be desirable
  - May provide better density estimation due to less mass on the edges
- Overall results don't differ substantially between transformed and untransformed data
- Baseline can be used to understand variability in percent overlap
- An intuitive threshold such as 66% (at least two-thirds) or 75% (at least three-quarters) could also be used

TRENDS OVER TIME

PIECE-WISE LINEAR REGRESSION TRENDS OVER TIME

- On an annual basis, evaluate longer term trend
- Piece-wise linear regression
  - $Y = \beta_o + \beta_1 X_1 + \beta_2 (X_1 - Base) X_2$ where
  - Y = monthly metric (e.g., mean MV duration)
    X1 = number of months since the start of baseline period
    Base = the number of months in the baseline period
    X2 = 0 if X1 ≤ Base and 1 if X1 > Base
- If $\beta_2 \neq 0$, trend has changed from the baseline

TRENDS IN MEAN MONTHLY MV OVER TIME



- Baseline period was stable
- Significant increase in 2020
- Picked up by piece-wise regression

Piece-wise linear regression evaluating change in temporal trend of monthly mean MV duration between the Baseline period (2017-2019) and Year 2020.

		Mean	
Predictors	*Estimates*	*CI*	*p*
(Intercept)	310.64	276.89 – 344.39	**<0.001**
Months	0.70	-0.80 – 2.19	0.355
Index * (Months - 36)	11.40	5.11 – 17.68	**0.001**
Observations	48		
R^2 / R^2 adjusted	0.457 / 0.432		

CONCLUSIONS/NEXT STEPS

- Still working to refine approach and criteria
- Objective is for evaluations to run automatically on set schedule, generating reports and sending alerts as necessary
- Working with IT and HAC team to convert to Python and integrate with other system components
- Develop drift monitoring procedures for other variable types
  - Binary, counts, time-to-event